

Zilei Shao

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EDUCATION

University of California, Los Angeles

Ph.D. in Computer Science | Advisor: [Guy Van den Broeck](#)

Los Angeles, CA

Sep. 2024 - Present

Harvey Mudd College

B.S. in Joint Major of Computer Science and Mathematics | GPA: 3.94/4.0

Claremont, CA

Aug. 2020 - May 2024

RESEARCH EXPERIENCE

StarAI Lab, Los Angeles, UCLA

Graduate Researcher Advised by Professor Guy Van den Broeck

May 2024 – Present

Los Angeles, CA

- Integrating **tractable probabilistic models** into modern neural architectures to improve the training efficiency, inference quality, and controllability of deep generative models.
- Developing **optimization** techniques for probabilistic and generative models, spanning parameter learning, low-variance gradient estimation, and differentiable routing in large-scale architectures.
- Exploring **tokenization** in LLMs as a **neurosymbolic** problem, showing that non-canonical tokenizations of the same input substantially change model behavior.

Uncertainty in Reinforcement Learning, L.A.C.E. Lab, Harvey Mudd College

Undergraduate Researcher Advised by Professor Erin Talvitie

Jun. 2022 – Jul. 2023

Claremont, CA

- Developed bounding-box inference to propagate uncertainty in environment, increasing the performance of a model-based reinforcement learning algorithm Model-based Value Expansion (MVE) by over 100%.

PUBLICATIONS

- [1] Ian Li*, **Zilei Shao***, Benjie Wang, Rose Yu, Guy Van den Broeck, Anji Liu. Breaking the Factorization Barrier in Diffusion Language Model. *In Proceedings of the 43rd International Conference on Machine Learning, 2026.* (ICML 2026) [PDF]
- [2] Heng Zhao, **Zilei Shao**, Guy Van den Broeck, Zhe Zeng. ProbMoE: Differentiable Probabilistic Routing for Mixture-of-Experts. *In Proceedings of the 43rd International Conference on Machine Learning, 2026.* (ICML 2026) [PDF]
- [3] Anji Liu, **Zilei Shao**, Guy Van den Broeck. Rethinking Probabilistic Circuit Parameter Learning. *In Proceedings of the 29th International Conference on Artificial Intelligence and Statistics, 2026.* (AISTATS 2026 Oral) [PDF]
- [4] **Zilei Shao**, Anji Liu, Guy Van den Broeck. Zero-Variance Gradients for Variational Autoencoders. *In NeurIPS 2025 Workshop on Structured Probabilistic Inference & Generative Modeling, 2025.* (Best Poster Award) [PDF]
- [5] Renato L. Geh*, **Zilei Shao***, Guy Van den Broeck. Adversarial Tokenization. *In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, 2025.* (ACL 2025 Main Conference) [Webpage, PDF]
- [6] Erin J. Talvitie, **Zilei Shao**, Huiying Li, Jinghan Hu, Jacob Boerma, Rory Zhao, Xintong Wang. Bounding-Box Inference for Error-Aware Model-Based Reinforcement Learning. *Reinforcement Learning Journal, vol. 1, no. 1, 2024.* (RLC 2024) [PDF]

PROFESSIONAL EXPERIENCE

Waymo LLC

Jun. 2025 – Sep. 2025

- Software Engineer Intern in Predictive Planner team.
- Improved motion prediction models by developing dynamic model capacities for more accurate agent forecasting.

TALKS

- [1] Token Dependencies as Vulnerability and Opportunity in Language Models, Zilei Shao. *Harvey Mudd College Summer CS Talk, 2026.*
- [2] Adversarial Tokenization, Zilei Shao. *Shang Data Lab at UCSD, April 2025.*
- [3] Range-to-Range Regression: Propagating Uncertainty in Environment Model, Zilei Shao, Erin Talvitie (advisor). *Harvey Mudd College Summer CS Talk, 2023.*

Introduction to Computer Science, UCLA Computer Science Summer Institute

Summer 2026

Lead Instructor | Faculty Supervisor: Todd Millstein

Los Angeles, CA

- Lead instructor for a three-week introductory Python course for high school students with no prior programming background; delivered daily three-hour lectures to a full cohort, designed the entire curriculum, and supervised a team of three teaching assistants.

Teaching Assistant

- CS161 Fundamentals of Artificial Intelligence, UCLA, 2025 Fall.
- CS5 Introduction to Computer Science, Harvey Mudd College, 2024 Spring, 2023 Fall.
- MATH179 Mathematics of Big Data, Harvey Mudd College, 2024 Spring.
- CS81 Computability and Logic, Harvey Mudd College, 2023 Spring.
- CS70 Data Structures and Program Development, Harvey Mudd College, 2022 Fall.
- CS181Y CS for Inquiry, Harvey Mudd College, 2022 Spring.
- CS60 Principles of Computer Science, Harvey Mudd College, 2021 Fall.